

Yuan Zhang

PH.D. CANDIDATE · RESEARCH INTERESTS: COMPUTATIONAL SOCIAL SCIENCE, AI SAFETY AND GOVERNANCE, MULTI-AGENT MODELLING

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Education

Ph.D. Communication Science

08/2022–05/2026

University of Zurich

Zurich, Switzerland

- International and Comparative Media Research Division, Department of Communication and Media Research (IKMZ)
- Quantitative Network Science Group, Department of Mathematical Modeling and Machine Learning (DM3L)
- Funded by the Swiss National Science Foundation (SNSF) project grant

Visiting Scholar

01/2025–06/2025

University of Pennsylvania

Philadelphia, PA, United States

- Visiting Scholar at Center for Information Networks and Democracy, Annenberg School for Communication

M.Sc. Social Data Science

10/2021–08/2022

University of Oxford

Oxford, United Kingdom

- Oxford Internet Institute
- GPA: Merit

M.Sc. Political Science and Political Economy

09/2018–11/2019

London School of Economics

London, United Kingdom

- Department of Government
- GPA: Distinction

B.Sc. Economics

09/2014–06/2018

Yunnan University

Kunming, China

- Department of Economics
- Average Score: 91/100, among the top 5%; GPA: 4.0/4.0

Research Experience

Ph.D. Thesis

08/2022–05/2026

University of Zurich

Zurich, Switzerland

- **Doctoral Committee:** Prof. Dr. Sandra González-Bailón (University of Pennsylvania), Prof. Dr. Jing Zeng (University of Zurich)
- **Advisors:** Prof. Dr. Frank Esser (University of Zurich), Prof. Dr. Laia Castro Herrero (University of Barcelona), Prof. Dr. Alexandre Bovet (University of Zurich)
- **Dissertation:** Computational Modeling of Online Political Confrontation: Real-World Complexity and Methodological Advances
- **Project 1: Automated Political Incivility Detection and Interpretation** (08/2022–10/2025)
 - *Methods:* Active Learning, Human-in-the-Loop, Linguistic Graph, LLM Automatic Prompting Optimization (APO)
 - Involved in designing an active learning pipeline with human-in-the-loop validation to address data imbalance challenge in automatic classification of online uncivil discourse
 - Designed LinGO, a linguistic graph-based LLM automatic prompting optimization framework, enabling LLMs to interpret fine-grained incivility intentions (including explicit, implicit, reporting, amplifying, countering, escalating)
 - *Output:* Published at AAAI ICWSM 2025 (see Paper 3); additional manuscript under review (see Paper 4)
- **Project 2: Quantifying Multi-Scale Online Political Fragmentation** (08/2022–10/2025)
 - *Methods:* Graph Mining, Multi-Scale Community Detection, Random Walks, Effective Branching Factor
 - Detected hierarchical communities among online political influencers using random walk-informed multi-scale community detection, and developed indices to measure ordinary users' multilevel selective exposure patterns
 - Created country-level online political fragmentation measure based on effective branching factor
 - *Output:* Published at PLOS ONE (see Paper 1); additional manuscript under review (see Paper 5)

- SNSF-funded project: *From Uncivil Disagreement to Political Unrest? A cross-platform & cross-national analysis of the offline consequences of online incivility*
- Collected and maintained large-scale social media datasets (Twitter/X, Facebook, YouTube, Instagram) in PostgreSQL and SQLite
- Developed taxonomy of nine forms of political incivility and hybrid classification models (sentence transformers + LLMs) for automatic detection tasks
- Coordinated the launch of national surveys during political elections across three countries: Brazil, Spain, and the United States
- Built automated scrapers to collect public news article content from tracked user web browsing activities

Research Assistant

03/2022–06/2022

Oxford Internet Institute

Oxford, United Kingdom

- Actively participated in the discussion of socio-political theory in the Internet age in India and China with Prof. Ralph Schroeder
- Fine-tuned RoBERTa model on a self-collected Chinese corpus for populist content classification
- *Output*: Published in *Social Media + Society* (see Paper 2); contributed to discussions in the chapter “Online Contention: Populism and Anti-Globalism” in Ralph Schroeder’s book: *Online Politics in India and China: Social Theory Beyond East and West*, Oxford University Press: <https://academic.oup.com/book/62734>.

Non-Academic Work Experience

AI Strategy Consultant

06/2020–08/2021

China Mobile (Xiong’an)
Industrial Research Institute

Beijing, China

- Conducted macroeconomic and policy analyses and generated both qualitative and data-driven insights to support smart city development strategies
- Developed 5G+AI governance solutions in collaboration with engineers and policy makers
- Led a team in producing and publishing research reports on the telecommunications and AI industries

Teaching Experience

AI and Political Discourse: Measuring Incivility with LLMs

FALL 2025

University of Zurich

Zurich, Switzerland

- Instructor, Bachelor seminar

Transformation of Political Information Environments

SPRING 2024

University of Zurich

Zurich, Switzerland

- Co-instructor, Master research seminar

Publications and Work-in-Progress

JOURNAL ARTICLES

1. **Zhang, Y.**, Castro, L., Esser, F., & Bovet, A. (2025). *A multilevel network approach to revealing patterns of online political selective exposure*. *PLOS ONE*, 20(9), e0332663. <https://doi.org/10.1371/journal.pone.0332663>.
Also presented at International School and Conference on Network Science (NetSci) 2024, Québec.
2. **Zhang, Y.**, & Schroeder, R. (2024). *“It’s All About US vs THEM!”: Comparing Chinese Populist Discourses on Weibo and Twitter*. *Social Media + Society*, 10(1), 20563051241229659.
Also presented at International Conference on Computational Social Science (IC2S2) 2023, Copenhagen.

CONFERENCE PAPERS

3. **Zhang, Y.**, Amsler, M., Herrero, L. C., Esser, F., & Bovet, A. (2025, June). *Quantifying the Spread of Online Incivility in Brazilian Politics*. *Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)*, 19, 2241–2259.
Also presented at NetSci 2025, Maastricht (**Best Poster Award**); IC2S2 2025, Norrköping.

WORK-IN-PROGRESS / UNDER REVIEW

4. **Zhang, Y.** (2026). *Pluralistic Alignment in Civility Judgments: Comparing AI Standards with Public Perceptions*. *Manuscript in progress*.
5. **Zhang, Y.**, & Bertaglia, T. (2025). *LinGO: A Linguistic-Graph Prompting Optimization Framework for Intent Classification in Uncivil Discourse*. *Under review*.

6. **Zhang, Y.**, Herrero, L. C., Esser, F., & Bovet, A. (2025). *Measuring Structural Political Fragmentation*. Under review. Presented at International School and Conference on Network Science (NetSci) 2025, Maastricht.

Reviewing

- Journal of Statistical Mechanics: Theory and Experiment (JSTAT)
- Proceedings of the International AAAI Conference on Web and Social Media (ICWSM)
- PLOS ONE
- EPJ Data Science

Conferences and Talks

- (Oral) The 2026 Conference on Complex Systems (CCS), Binghamton, United States, 9–16 October 2026 (accepted).
- (Oral) International Conference on Computational Social Science (IC2S2), Norrköping, Sweden, 21–24 July 2025.
- (Oral) International AAAI Conference on Web and Social Media (ICWSM), Copenhagen, Denmark, 23–26 June 2025.
- (Oral) 75th Annual International Communication Association (ICA) Conference, Denver, United States, 12–16 June 2025.
- (Oral+Poster) International School and Conference on Network Science (NetSci), Maastricht, the Netherlands, 2–6 June 2025.
- *Revealing multilevel political selective exposure, echo chambers, and polarization on social media & Integration of Theory and Methodology*, NetPlace Seminar, online, 14 November 2024.
- (Oral) International School and Conference on Network Science (NetSci), Québec City, Canada, 16–21 June 2024.
- (Oral) ECREA Political Communication Section Conference, Berlin, Germany, 31 August–1 September 2023.
- (Oral) International Conference on Computational Social Science (IC2S2), Copenhagen, Denmark, 17–20 July 2023.

Awards, Grants, and Honors

- Best Poster Award, NetSci Conference, Maastricht, Netherlands (2025)
- Swiss National Science Foundation (SNSF) Mobility Grant for Exchange Scholar (CHF 20,000) (2024)
- Guoyu Fang Scholarship, Yunnan University (2017)
- Provincial First Prize, “Challenge Cup” National Academic Competition (Team Leader) (2017)
- Provincial Government Scholarship, Yunnan Province (2016)
- National Scholarship, Ministry of Education of the People’s Republic of China (2015)

Additional Activities

- Summer Institute in Computational Social Science, University of Edinburgh, United Kingdom (2023)
- Summer School of the Standing Group on Political Communication, University of Amsterdam, Netherlands (2023)
- Project Management Intern, China Foundation for Poverty Alleviation (CFPA) (2018)
- Leader, News and Media Center, Yunnan University (2017)
- Volunteer Teacher, Primary School in Bali, taught a class of approximately 30 students (2016)

Technical Skills

- **Programming & Software:** Python, SQL, PyTorch, TensorFlow, Docker, GitHub, PostgreSQL, SQLite, LangChain, LangGraph, LLaMA Factory
- **Statistical Analysis:** R, Stata, MATLAB
- **ML Methods:** Transfer Learning, Large Language Models, Prompting Engineering/Optimization, LLM Fine-tuning, AI Agents
- **Research Methods:** Network Analysis, Statistical Modeling, Data Mining, Causal Inference, Time-series Forecasting

Miscellaneous

- **Languages:** Mandarin (native), English (C1), German (B1), Dutch (A1)
- **Hobbies:** Oil painting, photography